Frozen Operational Tests of a Bounded Dirac Role Across Two Registered Local Quantum Updates

Douglas Clark

Manuscript draft — July 2026

Abstract

The one-dimensional Dirac quantum cellular automaton and continuum limit are established. We ask whether a bounded Dirac operational role can be tested prospectively across two separately registered local updates under frozen maps, interventions, resources, and verification rules. The target is the $1+1$-dimensional Dirac equation on a periodic domain, with inputs restricted to an 18-dimensional low-momentum sector and horizon $T = 2$. We compare an ordered product walk and the established symmetry-derived direct Dirac QCA, with an explicit two-layer hard-core-qubit localization. Each update used four frozen development scenarios and two indivisible, single-use confirmation scenarios. Primary dynamics used 256-bit outward-rounded ball arithmetic; an algorithmically separate 110-decimal-digit implementation reconstructed all four counted v0.4 stage ledgers, which recorded positive classifications. Uniform evolution, witness, causal, symmetry, and analytic leakage evidence remains intact. A post-result hostile audit found that both implementations shared a factor-of-two empirical zone-edge predicate error that selected an empty sector. Those values and pass flags, plus a rough-potential control, are withdrawn. The separate pre-outcome analytic bound used the correct boundary and remains below 1.84×10^{-164} against a 10^{-24} gate. The largest certified scenario distances were 0.03000726 and 0.002759767 for the product and direct updates. The free one-excitation blocks are exactly related by a half-cell time-frame conjugacy after mass-angle remapping, and their declared resource counts depend on nonminimal gate-fusion conventions. The intact evidence therefore supports qualified constructive sufficiency for the shared bounded role core across the two specifications, not the complete originally registered direct role, algebraically independent dynamics, architecture-class breadth, whole-zone Dirac equivalence, a canonical fermionic lift, minimality, or ontology. An exact zone-edge opposite-mass pseudodoubler makes the global boundary explicit.

1 Introduction

Local quantum walks and quantum cellular automata (QCA) provide exact unitary models in which finite-range update rules approach relativistic wave equations. Product-form Dirac walks have rigorous observational convergence results [1]. A one-dimensional direct Dirac QCA was derived from homogeneity, locality, unitarity, parity, and time-reversal requirements, and was compared with continuum Dirac evolution through unitary channel discrimination [2, 3]. Zitterbewegung and scattering from a position-dependent potential are likewise established for that direct family [4]. The existence of the direct automaton, its low-momentum Dirac limit, and those dynamical phenomena are therefore not the novelty of the present work.

The open methodological question addressed here is narrower and, in a different sense, more demanding. Suppose a target theory is represented not by visual resemblance or one asymptotic expansion, but by a frozen collection of operational roles: a uniform evolution bound on a declared input sector, fixed encoding and intervention maps, fixed measurement witnesses, causal and symmetry requirements, explicit resources, and independent numerical reconstruction. Can two separately specified finite local updates satisfy the same bounded role after each test has been fixed and its confirmation cases withheld?

This formulation separates three questions that are often collapsed. First, an *analytic construction* can establish exact locality, unitarity, symmetry, or a formal continuum expansion. Second, a *registered operational test* can establish that a particular finite construction meets declared error and intervention criteria on a bounded domain. Third, a *structural breadth claim* asks whether mechanisms that remain distinct under a declared equivalence relation pass the role. Passing one construction does not answer the third question, while two frozen specifications that are related by a change of frame do not establish an architecture-wide theorem.

We implement the operational comparison for two radius-one updates. The first is an ordered product of a scalar phase, an on-site mass mixer, and a two-layer conditional transport circuit. The second uses the established direct Dirac QCA one-particle law, but declares and audits a full-space hard-core-qubit localization comprising an edge scatter and an on-site routing SWAP. As registered, the updates have different gate partitions and finite-step signatures: the product walk has a leading Lie–Trotter ordering error, while the direct free update has an even-in-step effective-generator correction. Their one-excitation blocks are nevertheless related by an exact half-cell time-frame conjugacy and gate fusion, which we make explicit below. We therefore claim only enumerative coverage at the registered-specification level, not two algebraically independent dynamical families or an architecture class.

The evidence was produced sequentially: the product result completed before the direct follow-on was frozen. This is not a jointly frozen two-arm trial. It is prospective within each registration. For each update, candidate law, target, maps, scenarios, states, metrics, thresholds, numerical precision, and stopping rule were content-sealed before registered outcomes. Development was evaluated as one indivisible batch. Only a committed development pass accepted by an algorithmically separate reconstruction could open a fresh, indivisible, single-use confirmation batch. Confirmation is not treated as a large statistical sample: the primary certificate is uniform over a fixed 18-dimensional input sector, and the held-out intervention schedules provide a non-algebraically-implied test of the frozen construction.

The resulting contribution is fourfold:

1. one common bounded Dirac role is made executable through fixed encoding, intervention, and readout maps rather than trajectory inspection;
2. two separately specified local updates are tested under separately frozen, realization-appropriate convergence gates;
3. declared circuit clocks, causal support, arithmetic precision, and algorithmically separate reconstruction are reported as part of the evidence; and
4. the registered evidence is paired with explicit time-frame, compilation, global-spectrum, lift, and post-result audit boundaries.

The four counted v0.4 stage ledgers recorded positive development and single-use confirmation classifications. A post-result audit withdrew one shared empirical π-sector reducer from both primary and replay evidence while preserving its separate pre-outcome analytic bound. The supported conclusion is consequently one of *qualified constructive sufficiency for the intact shared bounded core across two registered update specifications*. It is not the first operational comparison between a QCA and the Dirac theory; it is a sequential, separately prospective, multi-criterion comparison across two finite updates with explicit resource and provenance controls.

2 A frozen bounded Dirac operational role

2.1 Target and registered domain

In units $\hbar = c = 1$, the target spinor on the periodic domain $x \in [0, L)$ obeys

$$\mathrm{i}\,\partial_t \psi(x,t) = H_T(t)\psi(x,t), \qquad H_T(t) = -\mathrm{i}\sigma_z \partial_x + m(t)\sigma_x + V(x,t) I_2, \tag{1}$$

with $L = 16$. With the convention

$$\psi(x) = L^{-1/2} \sum_{r \in \mathbb{Z}} \widehat{\psi}_r e^{\mathrm{i} k_r x}, \qquad k_r = 2\pi r / L, \tag{2}$$

the registered input sector is

$$\mathcal{K}_{\mathrm{in}} = \mathrm{span}\{|r, s\rangle : -4 \le r \le 4,\ s \in \{R, L\}\}. \tag{3}$$

Thus $\dim \mathcal{K}_{\mathrm{in}} = 18$ and $|k| \le \pi/2$. Every registered state is normalized and supported entirely in $\mathcal{K}_{\mathrm{in}}$. Both registrations use physical horizon $T = 2$ and exact rational schedule boundaries. They deliberately use different lattice ladders and observation grids: the product walk uses $N = 256, 512, 1024$ and times $1/2, 1, 3/2, 2$; the direct QCA uses $N = 320, 640, 1280$ and times $2/5, 4/5, 6/5, 8/5, 2$. The distinction prevents a failed update from inheriting the successful update's numerical registration.

The target is finite-domain continuum Dirac evolution, not the lattice model itself. Potential schedules are piecewise constant in time and formula-defined in space. The product registration retains target Fourier modes $|r| \le 64$; the direct registration retains $|r| \le 72$. Outward analytic tails charge the amplitude that can leave these target cutoffs. No registered input contains a future target trajectory.

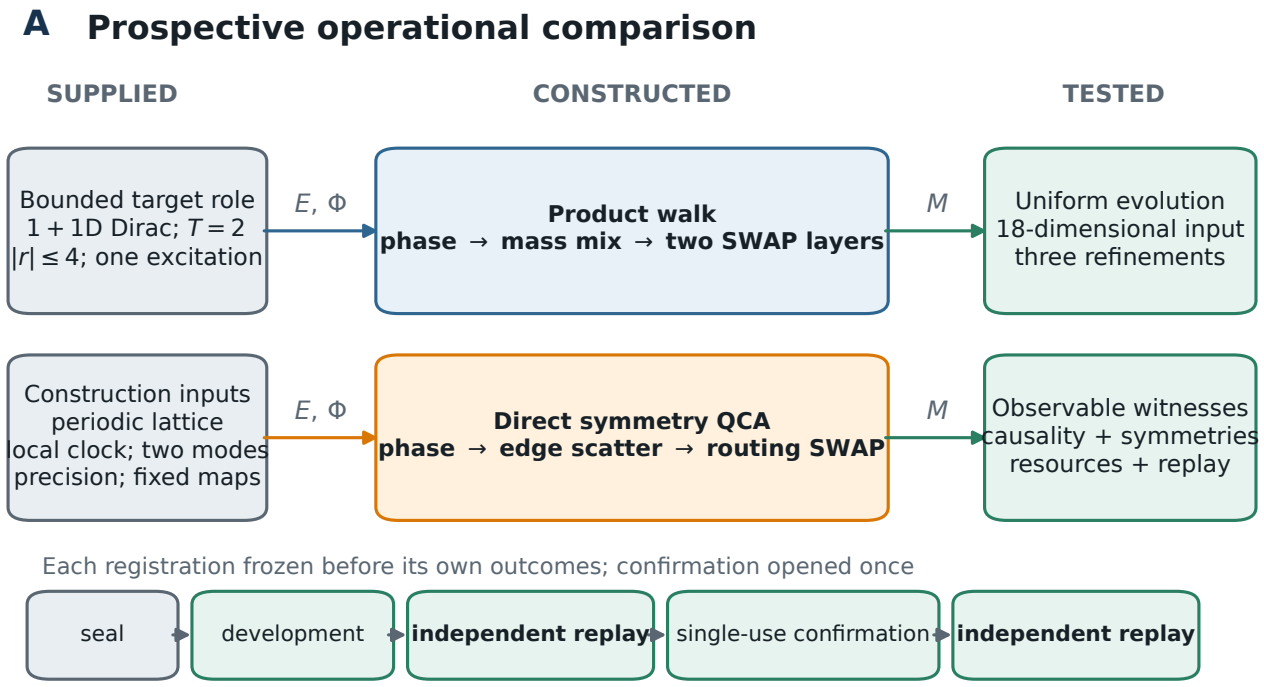


Figure 1: **Frozen operational-comparison contract (CONCEPTUAL).** The bounded target, quantum substrate, clock, constants, precision, maps, interventions, and gates are supplied. The constructed objects are two separately registered local updates. Both are tested with a uniform evolution bound, observable witnesses, structural requirements, resource accounting, and an algorithmically separate replay, subject to the shared π-reducer exception in Sec. 4.5. The arrows denote operational maps, not a derivation of the Dirac equation or quantum theory.

2.2 Fixed maps

The role is defined by three maps held fixed within each registration (Fig. 1). The encoding E_N copies coefficients at $r = -4, \ldots, 4$ into the identically labelled normalized discrete Fourier modes of the N-cell construction and zeros all other modes. It is an isometry; interpolation, aliasing, and outcome-dependent reconstruction are forbidden.

The intervention map Φ_N sends the exact mass schedule and scalar potential formula to the candidate's local gate parameters at each macrostep. It may not alter the encoding, readout, lattice size, transport law, or error threshold. The product walk receives mass angle Δm_q; the direct QCA receives $\mu_q = \Delta m_q$ and $n_q = \sqrt{1 - \mu_q^2}$. In either registration, an active scalar potential is a separately charged leading number phase sampled at the frozen macrostep convention.

The readout M_N applies the same Born-rule harness to target and candidate. It returns the position-grid distribution, momentum distribution, total directional current $\langle \sigma_z \rangle$, signed local-current vector, norm, and fidelity. The direct protocol additionally registered an empirical probability gate in the zone-edge sector $|q| \geq 3\pi/4$; a post-result audit found that its implemented predicate selected an empty sector, so those values and pass flags are withdrawn (Sec. 4.5). Zitterbewegung states retain the complete macrostep current series rather than a fitted frequency, amplitude, or phase.

For normalized candidate and embedded target states ψ_C and ψ_T on the same N-cell grid, define position and momentum probabilities $p_j^{C,T}$ and $\widehat{p}_r^{C,T}$, local current $j_j^{C,T} = |R_j^{C,T}|^2 - |L_j^{C,T}|^2$, and total current $J^{C,T} = \sum_j j_j^{C,T}$.

The six shared witness discrepancies are

$$\begin{aligned}
\delta_x &= \tfrac{1}{2}\sum_j |p_j^C - p_j^T|, & \delta_k &= \tfrac{1}{2}\sum_r |\widehat{p}_r^C - \widehat{p}_r^T|,\\
\delta_J &= |J^C - J^T|, & \delta_j &= \sum_j |j_j^C - j_j^T|,\\
\delta_F &= \max\{0, 1 - |\langle\psi_C, \psi_T\rangle|^2\}, & \delta_Z &= \max_{t\in\mathcal{O}_a} |J^C(t) - J^T(t)|.
\end{aligned} \tag{4}$$

Here $\mathcal{O}_a$ is the registered macrostep observation grid for a Zitterbewegung scenario. Every reported value is an outward upper enclosure of the corresponding expression.

2.3 Uniform comparison and witness hierarchy

For scenario a, observation time t, and lattice size N, let

$$D_N(a,t) = U_{C,N}[a](t)E_N - J_N U_{T,R}[a](t)P_{\mathrm{in}}, \tag{5}$$

where P_{in} projects onto $\mathcal{K}_{\mathrm{in}}$, $U_{T,R}$ is the independently constructed truncated target evolution, and J_N embeds retained target modes into the candidate grid. Candidate amplitude outside the retained target modes remains in the first term and therefore counts as error. The primary registered quantity is

$$d_N(a,t) = \mathrm{upper}(\|D_N(a,t)\|_2) + \beta_{\mathrm{tail}}(a,t) + \beta_{\mathrm{num}}(a,t,N), \tag{6}$$

an outward upper bound rather than a midpoint estimate. Each scenario is judged by the maximum over its registered observation times.

This uniform operator comparison is the primary gate because it applies to every normalized input in $\mathcal{K}_{\mathrm{in}}$, including states not explicitly plotted. The state witnesses are nevertheless essential. Four formula-defined state families probe positive energy, negative energy, coherent interference, and positive/negative-energy Zitterbewegung. They test whether the uniform bound translates into the registered observables and whether coherence was preserved rather than replaced by a classical mixture.

The final layer contains structural gates that are not reducible to the operator plot: exact radius-one causal support before wrap, attainment of both massless cone boundaries, unitarity and numerical norm preservation, parity, time-reversal role recovery, implementation integrity, resource accounting, and agreement with the separate reconstruction. The replay shared the π-predicate defect but remains separate for the other numerical families. The historical classifier marked a scenario as passing only if every implemented applicable layer passed. The later semantic audit shows why this condition is necessary but not sufficient: an empty-set reducer can return a passing value without implementing the specified layer. Apart from that disclosed exception, the hierarchy prevents a visually persuasive state trajectory from compensating for a failed uniform, causal, or provenance criterion.

The tested discrete symmetries are explicit. With $\mathcal{K}$ denoting componentwise complex conjugation,

$$(\Pi\psi)(x_j) = \sigma_x\psi(-x_j \bmod L), \qquad \Theta = \sigma_x\mathcal{K}. \tag{7}$$

For a terminal candidate map U_a, the parity relation is $\Pi U_a \Pi = U_{a^\Pi}$, where a^Π reflects the scalar potential, and the time-reversal-role relation is $\Theta U_a \Theta^{-1} = U_{a^{\mathrm{rev}}}^\dagger$, where a^{rev} reverses the frozen schedule. The registered residuals are spectral norms of these differences restricted to the input-basis columns; active leading potential phases may leave a prospectively gated splitting residual rather than exact equality.

3 Two separately registered local update realizations

Both candidates live on an oriented periodic cycle of N cells with directional labels R and L. Registered states occupy the global one-excitation sector of dimension $2N$. The direct registration declares an explicit four-dimensional hard-core-qubit cell and full-space gates. The earlier product registration assigns local Hilbert dimension four and declares ordinary transport SWAPs, but specifies its mass mixer only on the one-excitation R/L block. Its vacuum and double-occupation phases are therefore not frozen. Both update laws are fully specified on the registered one-excitation sector, but the comparison cannot use the product collision lift to establish full-space update inequivalence.

3.1 Ordered product walk

The product candidate applies, from right to left,

$$U_{\mathrm{PW}}[q] = S\, C_m[q]\, P_V[q], \tag{8}$$

where

$$P_V[q]|j,s\rangle = e^{-\mathrm{i}\Delta V_q(x_j)}|j,s\rangle, \tag{9}$$
$$C_m[q] = e^{-\mathrm{i}\Delta m_q \sigma_x}, \tag{10}$$
$$S|j,R\rangle = |j+1,R\rangle, \qquad S|j,L\rangle = |j-1,L\rangle. \tag{11}$$

The transport S is not counted as a free relabelling. An on-site SWAP followed by a directed edge SWAP realizes the two conditional shifts. In the frozen, unoptimized decomposition, a free macrostep has depth three and gate-work $3N$; an active potential adds one leading phase layer and N gates. The free momentum block is

$$U_{\mathrm{PW}}(k;m) = e^{-\mathrm{i}\Delta k \sigma_z} e^{-\mathrm{i}\Delta m \sigma_x}. \tag{12}$$

Its leading global discrepancy is first order in Δ, reflecting the commutator of the ordered kinetic and mass factors [1].

Because the product mass mixer lacks frozen $|00\rangle$ and $|11\rangle$ actions, many full-space extensions share Eq. (12). The natural hard-core extension $\mathrm{diag}(1, e^{-\mathrm{i}\Delta m \sigma_x}, 1)$ would carry the same $+1$ double-occupation phase as the direct gates. No product multiparticle or collision-sector distinction is claimed.

3.2 Direct symmetry-derived QCA

For $q = k\Delta$, $\mu = m\Delta$, and $n = \sqrt{1-\mu^2}$, the established direct one-excitation law has momentum block

$$U_\mu(q) = \begin{pmatrix} n e^{-\mathrm{i}q} & -\mathrm{i}\mu \\ -\mathrm{i}\mu & n e^{\mathrm{i}q} \end{pmatrix}, \tag{13}$$

or real-space recurrence

$$\begin{aligned} R'(x) &= nR(x-1) - \mathrm{i}\mu L(x), \\ L'(x) &= -\mathrm{i}\mu R(x) + nL(x+1). \end{aligned} \tag{14}$$

The law and its symmetry origin are prior art [2, 3]. What must be declared for the present test is its finite full-space lift, circuit clock, and collision boundary.

On edge subcells (L_x, R_{x-1}), in basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, define

$$b_\mu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -\mathrm{i}\mu & n & 0 \\ 0 & n & -\mathrm{i}\mu & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{15}$$

All disjoint edge gates form B_μ; ordinary on-site SWAPs form A. With an optional leading number phase P_V, the physical macrostep is

$$U_{\mathrm{D}}[q] = A\, B_{\mu_q}\, P_V[q]. \tag{16}$$

The one-excitation restriction of Eq. (16) is exactly Eq. (14), coefficient by coefficient. Each edge gate is unitary because its one-excitation block is $nX - \mathrm{i}\mu I$ and $n^2 + \mu^2 = 1$; the vacuum and double-occupation blocks are identities. The two partition layers are homogeneous and disjoint within a layer. In its frozen decomposition, one macrostep has radius one, free depth two, free gate-work $2N$, and no ancilla.

This lift is an ordinary hard-core occupation-qubit QCA. It fixes $|11\rangle$ with phase $+1$. Canonical fermionic second quantization of the same one-particle two-mode block would apply its determinant, -1, in the double-occupation sector. The two lifts agree on vacuum and one excitation but need not agree on multiparticle scattering or spectrum [5, 6]. The difference is an explicit claim boundary, not a hidden implementation detail.

Table 1: Declared circuit and resource comparison. Depth and gate-work are frozen protocol-clock counts, not optimized compilation minima.

Coordinate	Product walk	Direct symmetry QCA
Free update	Ordered mass mix and transport	Single normalized QCA block
Declared localization	Mass mix; on-site and edge SWAPs	Edge scatter; on-site routing SWAP
Full-space scope	Mixer collision phase not frozen	Explicit hard-core +1 collision phase
Declared free / active depth	3 / 4 layers	2 / 3 layers
Declared free / active gate-work	$3N$ / $4N$	$2N$ / $3N$
Storage; ancillas	$2N$ mode labels; 0	$2N$ qubits; 0
Causal radius	one cell per macrostep	one cell per macrostep
Leading free correction	first-order global, Lie–Trotter	second-order global, symmetric time frame
Registered refinements	$N = 256, 512, 1024$	$N = 320, 640, 1280$

3.3 Exact time-frame relation and different registered circuits

The one-excitation blocks are algebraically close. With $K(q) = e^{-\mathrm{i}q\sigma_z}$, the explicit half-step $K(q)^{1/2} := e^{-\mathrm{i}q\sigma_z/2}$, and $\theta_\mu = \arcsin\mu$, the direct block obeys

$$U_\mu(q) = K(q)^{1/2} e^{-\mathrm{i}\theta_\mu\sigma_x} K(q)^{1/2} = K(q)^{-1/2}\big[K(q)e^{-\mathrm{i}\theta_\mu\sigma_x}\big]\,K(q)^{1/2}. \tag{17}$$

Discrete-time quantum walks admit unitarily related stroboscopic time frames, conventionally represented by cyclic factor orderings $U' = FG$ and $U'' = GF$ [7]. Equation (17) is the model-specific instance here: the direct block is exactly a symmetric split and is conjugate to a product-walk block after the mass-angle remapping $\Delta m \mapsto \arcsin(m\Delta)$. This identity is a claim boundary, not a defect to conceal. The conjugating factors are chiral half-cell phases, not the integer-cell transport layers of the registered product circuit. The frozen specifications use different gate partitions, intervention parameterizations, and protocol clocks. The phrase *two registered updates* therefore means two declared one-excitation realizations under the fixed cell partition, encoding, intervention map, and macrostep accounting. It does not mean inequivalence under arbitrary momentum-dependent frame changes or circuit resynthesis.

The resource difference is equally convention-bound. In the product clock an adjacent mass mixer and SWAP can be fused as $Xe^{-\mathrm{i}\theta X} = \cos\theta\, X - \mathrm{i}\sin\theta\, I$, the same parameterized one-excitation form used by the direct edge scatter after the time-frame change. A library that treats arbitrary two-mode unitaries as primitives can therefore compress the product clock, while a SWAP-plus-rotation library would decompose the direct scatter. We report the sealed, nonminimal layer and gate counts because they make the executed protocols auditable, not as compilation-invariant complexity or a resource advantage.

On the principal small-step branch, the direct block's exact effective generator is

$$H_{\mathrm{eff}}(k,\Delta) = H_D(k) + \Delta^2 K_2(k,m) + O(\Delta^4), \qquad H_D(k) = k\sigma_z + m\sigma_x, \tag{18}$$

with

$$K_2(k,m) = -\frac{m^2 k}{3}\sigma_z + \frac{m(k^2+m^2)}{6}\sigma_x. \tag{19}$$

Thus one direct free step differs from $e^{-\mathrm{i}\Delta H_D}$ at order Δ^3, and its global free discrepancy is second order. Moreover, $U_{-\Delta}(k) = U_\Delta(k)^\dagger$ exactly and $H_{\mathrm{eff}}(k,-\Delta) = H_{\mathrm{eff}}(k,\Delta)$: the finite interaction clock leaves an even-in-step correction without a microscopic arrow of time. An active leading potential phase reintroduces a first-order Lie-product contribution, which is why the direct protocol has separate direct-only and potential-split gate families.

4 Prospective protocol and independent reconstruction

4.1 Separate registrations, common logic

The two studies share a target-domain and decision vocabulary but not a numerical registration. The *same role* means the same continuum target, input sector, map semantics, witness families, and causal, symmetry, and resource categories. Candidate law, lattice ladder, intervention cases, numerical tolerances, and analytic convergence

B

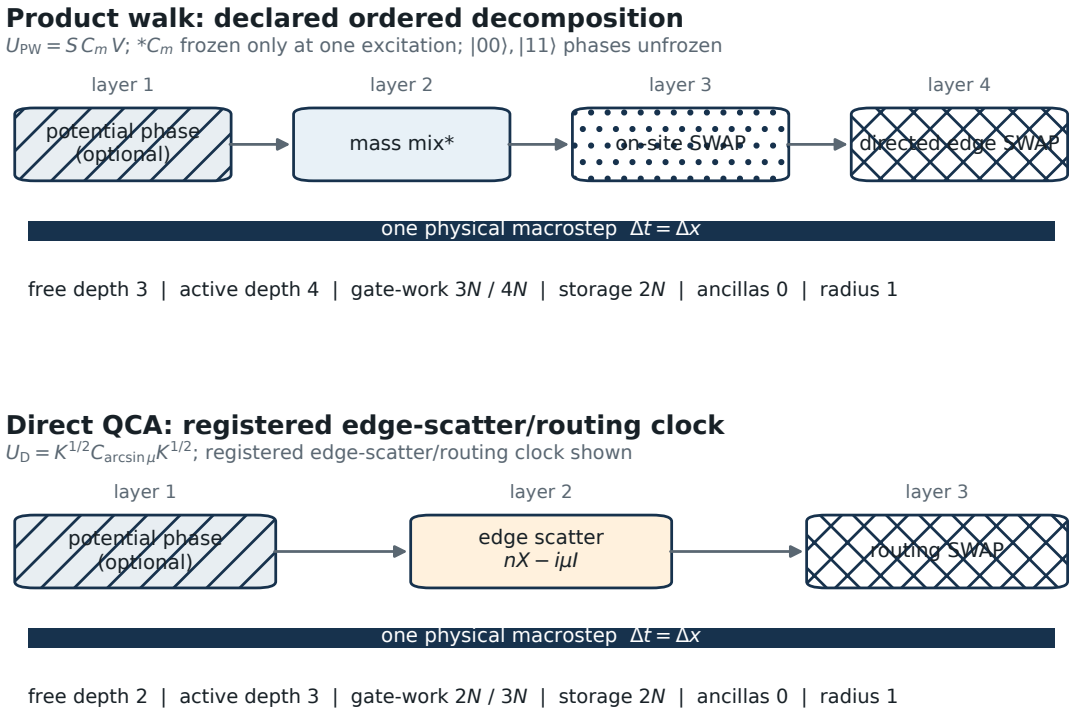


Figure 2: **Two registered local circuits under their declared clocks (ANALYTIC).** The product walk uses a mass mixer followed by two SWAP transport layers. The direct QCA uses a mass-bearing edge scatter followed by on-site routing. An active scalar potential contributes one leading phase layer. Counts are exact for the frozen decompositions but are nonminimal and change under gate fusion; they are not a hardware compilation or an efficiency comparison. At one excitation the direct block is exactly conjugate to a mass-remapped product block by the half-cell time-frame change in Eq. (17); the distinction shown here is the as-registered decomposition and clock.

order are realization-specific and were frozen separately. A content seal binds candidate and target code, state and scenario formulas, maps, partitions, reducers, result schema, tests, dependencies, precision, and stopping rule. Registered execution remains disabled until the seal is committed alone and identified by an annotated freeze tag. A stage-opening manifest is then committed alone. The stage runs as one indivisible batch and writes one no-clobber ledger, which is committed before inspection or replay. Confirmation can open only if the committed development ledger passes both its primary validator and the independent reconstruction. This sequence is shown at the foot of Fig. 1.

The product development partition contains free light-mass, free heavy-mass, upward mass-quench, and even-potential schedules. Its confirmation partition contains a downward mass quench and an odd-potential schedule. The direct development partition contains positive mass, negative mass, a sign-changing quench, and a harmonic-2 potential. Its confirmation partition contains a three-stage mass pulse and a harmonic-3 potential. Each partition uses four formula-defined state families. Product confirmation also includes a predeclared cross-witness of its confirmation states against one development schedule after the holdout is opened.

The confirmation schedules are single-use intervention holdouts, not draws from a population model. Because Eq. (6) is uniform on $\mathcal{K}_{\mathrm{in}}$, a finite witness-state split does not supply its logical force. The holdout tests whether the frozen map from a new exact intervention schedule to the unchanged candidate continues to satisfy all uniform, observable, structural, and replay gates.

4.2 Frozen convergence gates

For the product walk, every scenario must satisfy

$$\begin{aligned} &d_{256} \le 0.20, \qquad d_{512} \le 0.10, \qquad d_{1024} \le 0.05, \\ &d_{512} \le 0.60 d_{256} + 10^{-8}, \qquad d_{1024} \le 0.60 d_{512} + 10^{-8}. \end{aligned} \tag{20}$$

The scale was chosen before outcomes from the first-order product estimate $T\Delta|km|$, with room for the registered potential and certified tail.

The direct QCA uses its analytic finite-step signature. Direct-only scenarios must satisfy

$$\begin{aligned} &d_{320} \le 0.0040, \quad d_{640} \le 0.0012, \quad d_{1280} \le 0.0004, \\ &d_{640} \le 0.35 d_{320} + 10^{-10}, \quad d_{1280} \le 0.35 d_{640} + 10^{-10}. \end{aligned} \tag{21}$$

whereas potential-split scenarios must satisfy

$$\begin{aligned} &d_{320} \le 0.020, \quad d_{640} \le 0.011, \quad d_{1280} \le 0.006, \\ &d_{640} \le 0.62 d_{320} + 10^{-10}, \quad d_{1280} \le 0.62 d_{640} + 10^{-10}. \end{aligned} \tag{22}$$

An outcome-free 256-bit certificate bounded the complete direct one-step remainder, telescoped it through $T = 2$, charged the potential commutator, target-cutoff tail, and numerical budget, and allowed potential-driven states to leave the input band. It used intermediate mode bands $|r| \le 13, 17, 20$ across the three refinements and charged the remaining Dyson tail at every later step. Its worst harmonic-3 envelopes were 0.019662, 0.007805, and 0.002914, below Eq. (22). The certificate justified the gate scale; it did not compute any registered trajectory, ratio, witness, or classification.

At the finest refinement, every product witness must have position and momentum total-variation distance at most 0.05, total- and local-current error at most 0.10, infidelity at most 0.0025, and Zitterbewegung current-series error at most 0.10. The corresponding direct gates are 0.010, 0.010, 0.020, 0.020, 10^{-4}, and 0.020. Both registrations require norm residual at most 10^{-12} and independent state disagreement at most 10^{-11}. The direct potential scenarios also specified candidate π-sector probability at most 10^{-24} in addition to their analytic leakage tails; the empirical implementation of that requirement is the exception documented in Sec. 4.5.

4.3 Numerical and algorithmic separation

Primary registered dynamics use `python-flint 0.9.0` Arb/Acb arithmetic at 256 bits with outward enclosures; binary64 values are forbidden from the registered dynamics. The product primary evolves the candidate in real space and the target in a retained Fourier basis. Its independent route reconstructs candidate gates and target evolution at 110 decimal digits with `mpmath 1.3.0`.

For the direct update, separation is more explicit. The primary candidate applies the leading phase and recurrence in real space; the primary target uses certified free blocks and spectral exponentials. The independent free candidate builds separate 2×2 momentum blocks. Its potential candidate constructs sparse phase, edge-scatter, and routing operators. The independent target rebuilds the harmonic generator and applies its exponential through a separate high-precision route. Candidate and verifier do not share a candidate constructor, and target and verifier do not share a target constructor. This is an algorithmically separate reconstruction by the same project, not an external replication.

4.4 Infrastructure failures preserved rather than hidden

The direct experiment required four separately sealed lifecycle lineages. Version 0.1 stopped before ledger construction when the independent norm routine used a real-symmetric eigensolver on a complex Hermitian Gram matrix. No partial ledger was written or inspected. Version 0.2 wrote a primary-passing development ledger, but replay found that a Fourier basis had been constructed at ambient 53-bit precision before entry into the registered 256-bit context. The certified upper bound was rigorous but failed the frozen 10^{-65} replay tightness allowance. Confirmation remained unopened.

Version 0.3 repaired precision ownership and independently reconstructed all potential-free development scenarios, but its maximum-Gram-row-sum norm bound became materially loose after the potential coupled registered momentum columns. It again stopped before confirmation. Version 0.4 replaced only this norm-bound

reducer with a rigorous finite-precision similarity transform and Gershgorin eigenvalue enclosure. Approximate eigenvectors choose a basis but carry no proof; the certified inverse, transform, and discs do. Candidate, target, maps, states, scenarios, partitions, thresholds, controls, and stopping rules did not change across these infrastructure repairs. The v0.2 and v0.3 primary classifications are not counted as verified results. Their failure records demonstrate why a second implementation must test certified enclosures, not merely reproduce floating-point midpoints.

4.5 Post-result semantic audit of the zone-edge reducer

A hostile-reader audit after the v0.4 result found a shared factor-of-two error in one reducer. The specification defines $|q| \geq 3\pi/4$ with $q = 2\pi r/N$, hence

$$8|r| \geq 3N. \tag{23}$$

The primary reducer, independent verifier, and non-gating rough-potential control instead used $4|r| \geq 3N$. Since every discrete mode has $|r| \leq N/2$, that predicate selects an empty set. All serialized empirical π-probabilities and the rough-control value zero are therefore vacuous and withdrawn. The replay duplicated the predicate and is not independent evidence for this metric.

The pre-outcome analytic leakage routine is separate and uses the correct coarsest boundary $3N/8 = 120$. Its serialized ledger values are approximately 1.62×10^{-261} for the harmonic-2 schedule and 1.83×10^{-164} for harmonic 3, below the frozen 10^{-24} gate. Thus the low-band leakage threshold remains supported analytically, while the claim that its additional empirical reducer was executed correctly is withdrawn. No registered ledger is altered. An audit-only recomputation of the disclosed non-gating rough control with Eq. (23) gives 3.4860×10^{-8}, confirming the defect but neither repairing nor broadening confirmation. The serialized ledger classification field remains a content-addressed historical fact; this manuscript reports it only with this qualification.

5 Registered results and post-result qualification

Both registrations' v0.4 development and confirmation ledgers record positive classifications. The product walk passed all four development scenarios, all 16 development witness groups, both confirmation scenarios, all eight confirmation witness groups, and all four confirmation-state cross-witness groups. The direct QCA ledger classifier marked four of four development scenarios with 16 of 16 witness groups and two of two confirmation scenarios with eight of eight witness groups as passing. Each committed development and confirmation ledger was accepted by the algorithmically separate reconstruction as implemented. The direct π-sector predicate was later found to be empty in both primary and replay; its measured values, associated pass flags, and one rough-potential control are withdrawn (Sec. 4.5). The direct scenario counts are therefore historical ledger classifications, not a claim that every intended registered gate executed correctly. No threshold or registered scientific input was changed after confirmation opening.

5.1 Certified convergence

Figure 3 plots every intact registered scenario-level d_N series; the withdrawn empirical π reducer is a separate metric and does not appear in this panel. For the product walk, the largest certified upper bounds across scenarios were 0.0300072572, 0.0149954636, and 0.00749669028 at $N = 256, 512, 1024$. The worst successive ratios were 0.500088317 and 0.500066986, below the frozen 0.60 gate. The curves show the registered first-order pattern without using a post-outcome slope fit as a criterion.

For the direct QCA, the largest direct-only bounds were $5.60191118 \times 10^{-4}$, $1.39960025 \times 10^{-4}$, and $3.49845248 \times 10^{-5}$. Their ratios lie between approximately 0.24984 and 0.24997, below the frozen 0.35 gate and consistent with the analytic second-order global discrepancy. The largest potential-split bounds were $2.75976665 \times 10^{-3}$, $1.37607978 \times 10^{-3}$, and $6.87867299 \times 10^{-4}$, with ratios between approximately 0.49444 and 0.49987, below 0.62. The active leading phase therefore produces the separately anticipated first-order global family rather than being hidden inside a direct-only convergence claim.

5.2 Observable and structural witnesses

The largest certified upper bound over development and confirmation for each update is reported in Table 2. Product maxima remained between about 11 and 249 times below their applicable gates; direct maxima remained

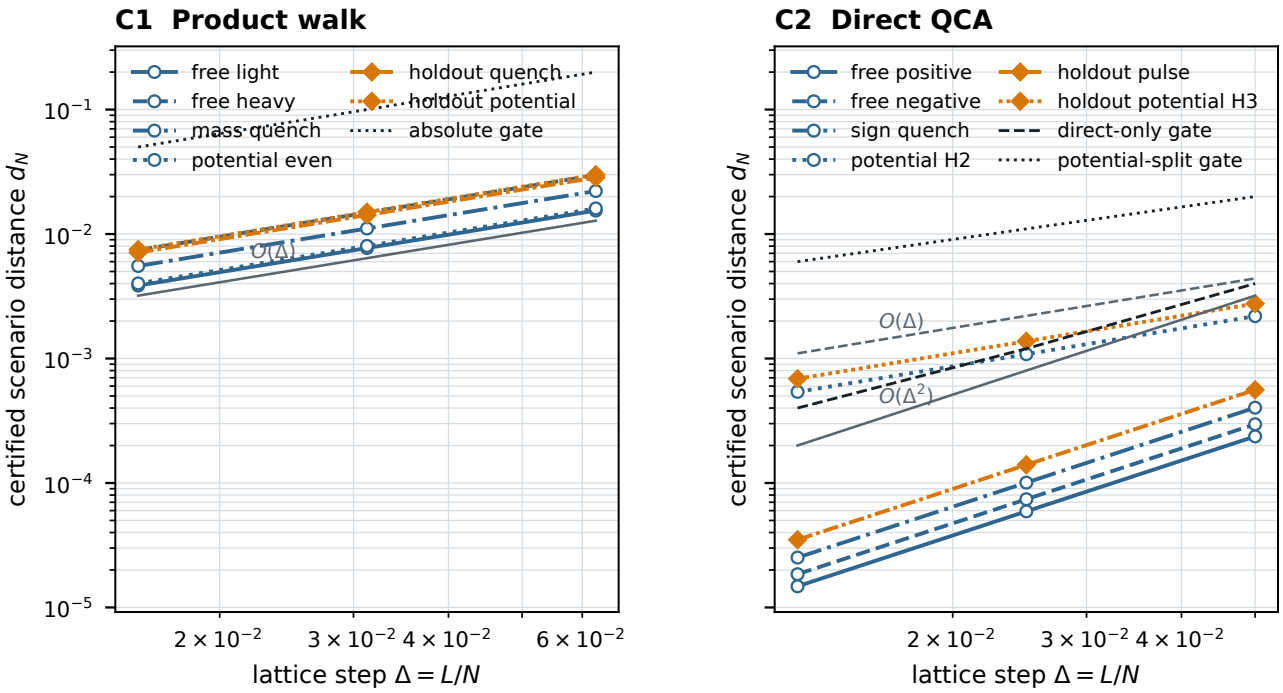


Figure 3: **Certified convergence of both frozen constructions (REGISTERED RESULT).** Each curve is the maximum over registered observation times of the serialized outward certified distance d_N for one scenario, including its analytic tail and numerical budgets. Open markers denote development; filled diamonds denote single-use confirmation. The product panel retains its first-order gates; the direct panel retains distinct direct-only and potential-split gates. Guide lines are analytic orders, not fits. The updates have different frozen refinement and observation grids, and values are not normalized by their gates.

between about 72 and 876 times below their gates, depending on the metric. These margins are descriptive, not a cross-registration score: the registrations have different steps, schedules, observation grids, and threshold rationales.

For the product walk, the largest witness norm residual was 3.39037×10^{-13}, the largest structural norm residual was 6.22285×10^{-13}, and the largest parity residual was 5.79543×10^{-14}, all within 10^{-12}. The largest fine time-reversal-role residual was 0.0134492 within its 0.10 gate and passed the first-order refinement requirement. Exact causal support and both massless cone boundaries passed. The primary and independent state vectors had serialized disagreement zero at the registered comparison precision.

For the direct QCA, the largest parity and unitarity residuals were 9.63632×10^{-74} and 1.22268×10^{-67}. Potential-split time-reversal-role residuals refined from 0.00550361 to 0.00137599 and passed their 0.62 ratio rule and 0.020 fine gate. All six scenarios passed the exact implementation-integrity and causal-support gates. The largest analytic π-tail bound was 1.83341×10^{-164}, below 10^{-24}. The empirical π values are not evidence because of Eq. (23). The largest state disagreement between primary and independent implementations on the remaining numerical families was 5.0626×10^{-17}, below 10^{-11}.

5.3 Declared resources and boundary controls

At the finest product refinement, a free $T = 2$ run uses 128 macrosteps, 384 elapsed elementary layers, and 393,216 gates. A registered active-potential run uses 448 layers and 458,752 gates. At the finest direct refinement, a free run uses 160 macrosteps, 320 layers, and 409,600 gates; the harmonic-2 and harmonic-3 potential schedules use 400 and 416 layers and 512,000 and 532,480 gates. These totals faithfully charge each sealed protocol clock. They are not an efficiency head-to-head because N, schedules, accuracy gates, and gate-fusion conventions differ. At equal N and macrostep count, the frozen decompositions assign free depth and work $3, 3N$ to the product walk and $2, 2N$ to the direct QCA. Section 3 shows why these nonminimal counts are not compilation-invariant and support no hardware or minimal-resource advantage.

Six direct-QCA controls ran only after the confirmation primary. Five remain informative: the exact opposite-

Table 2: Largest registered upper bounds for the six intact shared observable-witness families across both partitions. Gates are absolute and were frozen separately. The direct empirical π value and pass flag are withdrawn and are not included in this table.

Metric	Product max	Gate	Direct max	Gate
Position TV	0.00194047	0.05	9.8902×10^{-5}	0.010
Momentum TV	2.0061×10^{-4}	0.05	1.3971×10^{-4}	0.010
Total-current error	0.00844655	0.10	3.6737×10^{-5}	0.020
Local-current L_1	0.00844655	0.10	1.9639×10^{-4}	0.020
Infidelity	4.5070×10^{-5}	0.0025	3.1546×10^{-7}	0.0001
Zitterbewegung series L_∞	0.00881467	0.10	2.2840×10^{-5}	0.020

mass π-partner residual was 2.49490×10^{-111}; a high-q control had operator distance 0.00791313; the $|\mu| = 1$ endpoint had zero maximum group velocity; a $T = 12$ low-band control had distance 0.000821366; and the hard-core versus canonical fermionic double-occupation phases had distance 2. The rough-potential zero is withdrawn; the corrected audit-only diagnostic is 3.4860×10^{-8}. These non-gating controls characterize the boundary. They cannot repair a failed primary and do not broaden the result.

6 Discussion

6.1 What the qualified two-registration evidence adds

The central result is not that a lattice can approximate the Dirac equation. That is known. The intact evidence instead supports a shared core of one bounded operational role across two sequentially tested, separately prospective registered updates. The product walk satisfies that core through ordered local subdynamics; the direct QCA satisfies the same intact core through a normalized symmetry-derived block with a different localization, declared clock, and convergence signature, while its empirical zone-edge gate is withdrawn. The intact records support uniform propagator bounds, multiple observables, and causal and symmetry constraints for both updates. Algorithmically separate reconstruction agreed on those families while duplicating the withdrawn π-sector predicate. The result supports limited robustness at the registered-specification level, not independence of the underlying one-particle dynamics.

The coverage is only realization-level and enumerative: it covers these two tested registered updates under the equivalence convention stated around Eq. (17). The exact one-particle conjugacy rules out any stronger claim that the two laws define algebraically independent families. "Architecture independence" would require a defined architecture class and a theorem or sampling design over that class. Nor do the results support a primitive-class claim, because both constructions use the same quantum substrate, periodic lattice, Born readout, and externally specified clock and interventions. The stronger value of the second registration is therefore disciplined contrast, not unrestricted generalization.

6.2 Relation to established Dirac-QCA work

Bisio, D'Ariano, and Tosini already derive the direct automaton, its exact dispersion, low-mass/low-momentum Dirac regime, and a rigorous operational comparison framed as channel discrimination [2]. They also establish Zitterbewegung and potential scattering in the family [4]. Our contribution does not supersede those results. It embeds the established law as the second side of a sequential two-registration study with a common role core, realization-specific thresholds, an explicit direct-QCA full-space lift, declared protocol clocks, precision accounting, single-use confirmation, content-addressed ledgers, algorithmically separate reconstruction, and a disclosed post-result defect audit. The product full-space mass-mixer lift was not comparably frozen. That bounded package—not the generic idea of an operational comparison, and not architecture-class breadth—is the novelty claim.

6.3 Global spectrum: a necessary boundary

The direct QCA has exact principal band angle

$$\omega(q, \mu) = \arccos\Big(\sqrt{1 - \mu^2}\cos q\Big), \qquad q \in [-\pi, \pi). \tag{24}$$

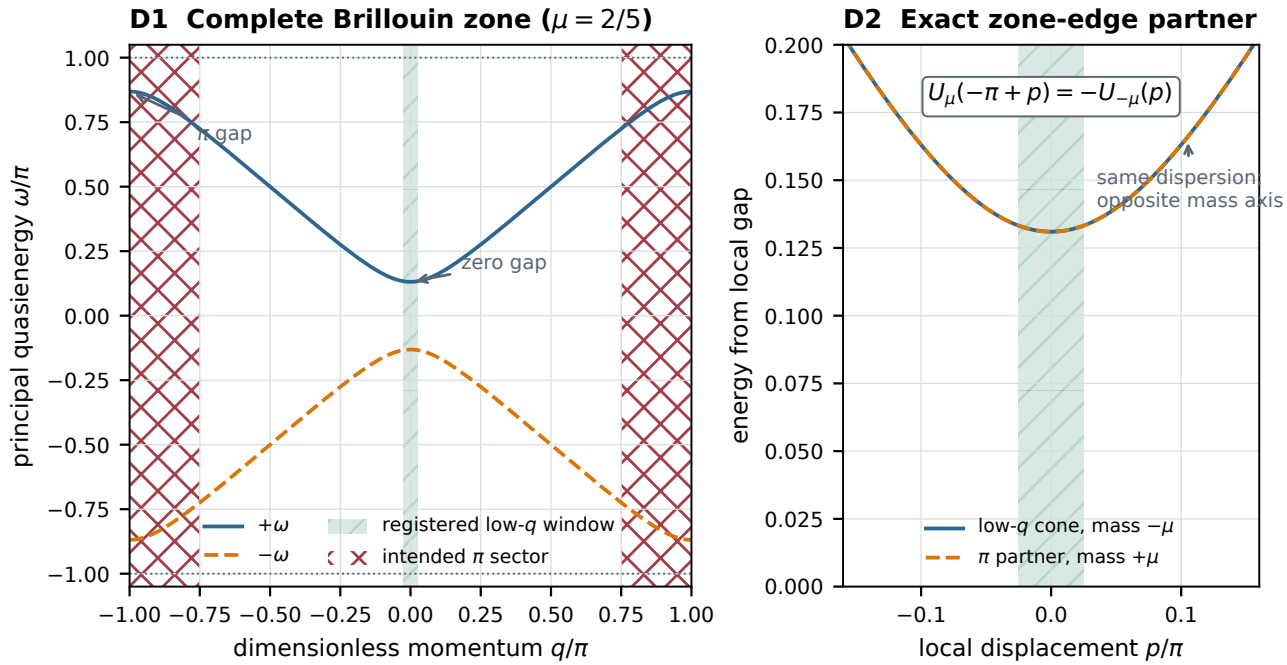


Figure 4: **Global spectrum and bounded claim domain (ANALYTIC; not a registered outcome).** For the disclosed illustrative value $\mu = 2/5$, the bands follow Eq. (24). The central strip is the coarsest registered input window $|q| \leq \pi/40$; finer windows are narrower. Hatching marks the intended π sector $|q| \geq 3\pi/4$ used by the valid analytic tail; the empirical reducer for that sector is withdrawn. The exact zone-edge identity exposes a high-quasienergy opposite-mass pseudodoubler.

For $0 < |\mu| < 1$, both the zero- and π-quasienergy circular gaps equal $2 \arcsin |\mu|$. There is no additional zero-quasienergy cone in $1+1$ dimensions and hence no conventional low-energy doubler. Independent analysis of the one-dimensional Dirac quantum walk likewise finds no isolated high-momentum/low-energy copy but does find a residual degeneracy under simultaneous quasienergy and momentum shifts to Brillouin-zone corners, motivating an explicit full-zone boundary even for a valid low-q continuum result [8]. For small zone-edge displacement p,

$$U_\mu(-\pi + p) = -U_{-\mu}(p). \tag{25}$$

After removal of the Floquet phase -1, the zone-edge mode has the same local law with the opposite mass sign. Following the distinction sharpened by Gupta and Short [9, 10], it is a high-quasienergy pseudodoubler. The registered low-q result is not invalidated: the input is separated from this sector, potential-generated leakage is charged by the pre-outcome analytic bound, and the primary metric compares propagators without asserting a global Hamiltonian logarithm. The intended empirical monitor is withdrawn under Sec. 4.5. The identity does, however, rule out a whole-zone or globally doubler-free interpretation.

At $\mu = 0$, both circular gaps close and the principal logarithm becomes branch-ambiguous at $q = -\pi$. Accordingly, the effective-generator expansion in Eq. (18) is confined to a branch-safe neighborhood of $q = 0$. This reconciles a bounded continuum role with the global Floquet spectrum and makes the limitation visible rather than relegating it to an unmonitored high-momentum sector.

6.4 Full-space lift and further tests

The hard-core/fermionic collision-phase difference is not visible to any one-excitation witness. It becomes material under second quantization, collisions, interactions, or a filled sea. Published causal fermionic and multiparticle QCA constructions show that those extensions require additional structure [5, 6]. They cannot be inferred from the present pass. A canonical fermionic or interacting follow-on should therefore receive a new search identifier, global-spectrum audit, vacuum definition, and prospective protocol.

The direct hard-core lift is explicit, but the product registration does not fix its mass mixer's vacuum and

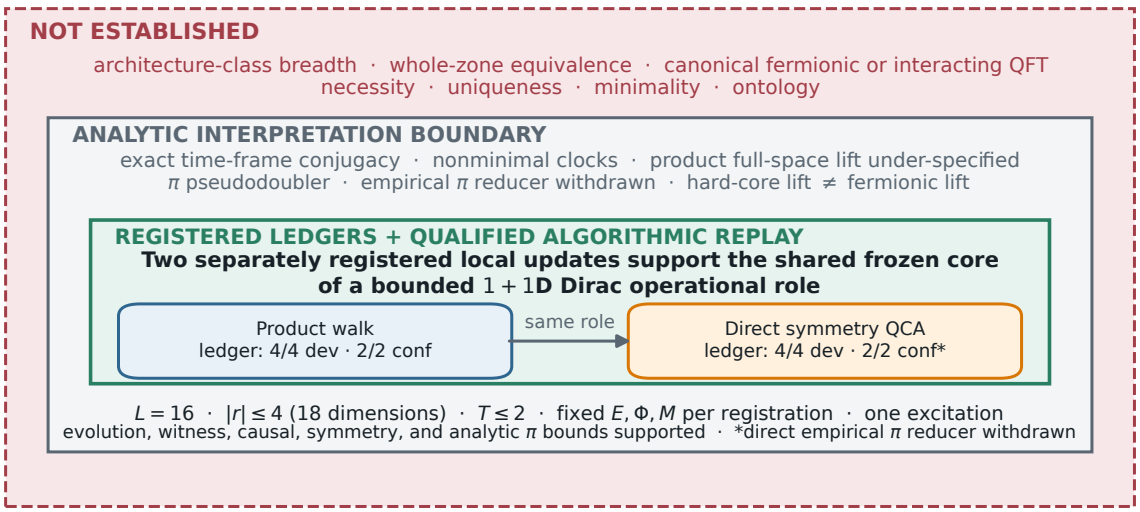


Figure 5: **Evidence and claim boundary (CONCEPTUAL synthesis).** The inner domain is the qualified registered result. The middle domain contains analytic constraints required to interpret it. The outer field lists claims not established. Inclusion denotes logical scope, not evidential probability. The withdrawn empirical π reducer is shown as an audit boundary rather than positive evidence.

double-occupation phases. The present one-excitation comparison therefore supplies no product-versus-direct collision claim. A follow-on that uses full-space behavior must freeze both lifts symmetrically.

The same separation applies to a differently localized circuit or a classical reversible lattice gas. A nominal Strang split is not automatically a third case because Eq. (17) already exposes an exact symmetric factorization of the direct block. A genuinely new case would need a predeclared circuit-equivalence criterion. Either direction could increase breadth, but neither changes the interpretation of the completed two-registration result. New science should begin only after this manuscript's claims and limitations are stable, under a separately registered search rather than as post-outcome robustness tuning.

7 Limitations and claim discipline

The result is deliberately bounded in several independent directions.

Finite physical domain. The target role covers $L = 16$, $T \leq 2$, exact registered mass and scalar potential schedules, input modes $|r| \leq 4$, and three refinements per update. It does not establish all-scale continuum equivalence, nonperiodic boundaries, arbitrary interventions, vector or gauge potentials, curved geometry, feedback, or long-time uniformity. Three refinements can test frozen ratio gates but cannot by themselves establish an unrestricted asymptotic theorem.

Single-excitation sector. All primary maps and outcomes concern one excitation. The direct circuit is unitary on the complete hard-core qubit Hilbert space, but its collision phase is not a canonical fermionic lift. Its empty hard-core vacuum is fixed, but no physically identified fermionic vacuum or Dirac sea, free-fermion field, multiparticle scattering, interaction, or particle creation is demonstrated.

Two registered updates, not an inequivalent architecture class. The qualified comparison covers a product walk and one direct QCA as registered one-excitation update specifications. Their one-excitation blocks

are exactly related by the half-cell time-frame conjugacy in Eq. (17), and their declared resource difference can be removed or reversed by changing the gate-fusion library. The product mass mixer's full-space collision phase was not frozen. The demonstrated difference is therefore only between the as-registered gate partitions, intervention parameterizations, and protocol clocks. The updates also share a quantum primitive, oriented cycle, Born readout, and externally specified clock. No equivalence class or distribution over architectures was defined, so there is no claim of architecture independence.

Constructive sufficiency only. The target Hamiltonian, lattice, quantum substrate, clock, constants, precision, maps, and intervention class are supplied construction inputs. The intact pass evidence shows that the declared local mechanisms are sufficient within the role. It does not show that they are necessary, unique, minimal, ontologically fundamental, or derivable from information alone. It also says nothing about a finite-classical-bit realization.

Verification independence. The replay uses a separately implemented numerical route with different candidate and target constructors and higher precision. It was produced by the same project and shares the frozen mathematical specification. We therefore call it algorithmically separate reconstruction or replay, not an external replication. A third-party clean-room reproduction remains valuable. The post-result audit also shows a limit of algorithmic separation: both implementations copied the same erroneous π-sector predicate, so replay is not semantic specification diversity.

Thresholds and record granularity. Absolute and ratio gates were analytically motivated and frozen, but any finite threshold encodes a chosen accuracy standard. Passing it is not exact continuum equality. Product ledgers serialize per-time non-Zitterbewegung witness observations; direct ledgers serialize their maxima and the full Zitterbewegung current series. The direct uniform operator records retain all five observation upper bounds, but their times are joined by canonical index to the target specification. This asymmetry does not affect registered maxima or classifications, but it prevents a perfectly symmetric per-time supplemental witness table.

Withdrawn empirical leakage reducer. The direct empirical π-sector reducer and rough-potential control are invalid because their predicate selects no modes. The pre-outcome analytic tail uses the correct boundary and remains below its frozen threshold, but the paper does not count the empirical zero or its duplicated replay as positive evidence. Corrected registered measurement would require a separately opened follow-on.

Figure 5 summarizes these exclusions. They are not generic disclaimers: analytic leakage accounting, the mass endpoint, long horizon, global spectrum, and collision-phase control make several of them executable failure boundaries. The withdrawn reducer demonstrates why semantic audits must complement byte-level replay.

8 Conclusion

Two separately prospective registrations tested an ordered product walk and the established direct Dirac QCA against a common, bounded, low-momentum single-excitation role. All four counted v0.4 stage ledgers recorded positive classifications, and an algorithmically separate implementation reconstructed the committed computations. The intact evidence supports qualified constructive sufficiency for the shared role core: uniform evolution bounds, observable witnesses, causal and symmetry requirements, and the correctly derived analytic zone-edge bound. It does not support the complete originally registered direct role, because the empirical zone-edge reducer and its pass flags are withdrawn.

The exact time-frame conjugacy also changes the interpretation of the second registration. It provides disciplined, separately frozen coverage of two declared update specifications and protocol clocks, not breadth across proven inequivalent architecture classes or a compilation-invariant resource advantage. The durable contribution is therefore the combination of an executable operational role, prospective stage separation, explicit locality and resource accounting, content-addressed evidence, algorithmically separate reconstruction, and an audit trail that preserves rather than conceals a post-result defect. Any genuinely broader architecture or corrected empirical zone-edge test belongs in a separately registered follow-on.

9 Reproducibility and content-addressed provenance

All plotted registered-result values and generated result tables derive from the four counted v0.4 JSON stage ledgers. The figure and table generator refuses to run if any ledger or registered specification hash differs from

Table 3: Content-addressed registered ledgers. The direct-QCA positive classifications are historical ledger fields qualified by the withdrawn empirical π-sector reducer. Full hashes are recorded in the generated manifest and experiment result summaries.

Architecture	Partition	Classification	Ledger SHA-256
Product walk	Development	`development_passed`	`c9f445358a7a...`
Product walk	Confirmation	`constructive_sufficiency_passed_frozen_dirac_sector`	`34c16fd95914...`
Direct QCA	Development	`development_passed`	`89d7c65a37bf...`
Direct QCA	Confirmation	`constructive_sufficiency_passed_direct_symmetry_qca_sector`	`be5a9f85770e...`

its recorded SHA-256. It parses serialized outward upper values, preserves scenario order, generates vector PDF and inspection PNG assets, and writes a sidecar manifest containing input, analytic-source, generator, and output hashes, tool versions, and the evidence class of every figure. No registered number is manually transcribed into a plot. Prose summaries are human-readable transcriptions and should be checked against the generated tables and ledgers.

The product history is linear from freeze commit `d942c114` through development opening `cfaaabe9`, development result `e5a5c4e9`, confirmation opening `0181ce0f`, confirmation result `c7e9a216`, and verified documentation `490dab6b`. The direct v0.4 history is linear from freeze `890962df` through development opening `f5f5d604`, development result `b96e70a1`, confirmation opening `6398124a`, and confirmation result `0795a6ef`. Each freeze, opening, and result commit introduces exactly its one registered artifact. Earlier direct lineages remain content-addressed and are not counted as positive evidence.

The complete local build is

```
make quantum-dirac-paper
```

which verifies ledger hashes, regenerates derived tables and all five figures, reproduces the post-result π-sector audit, runs the archived-result validator, and compiles this manuscript with `latexmk`. A full expensive reconstruction can be run separately through the checkpoint-specific direct-QCA verification commands documented with the result. The registered primary environment records `python-flint 0.9.0`, 256-bit Arb/Acb precision, and outward ball enclosures. The replay records `mpmath 1.3.0` at 110 decimal digits. The manuscript plotting layer uses pinned NumPy and Matplotlib versions only after ledger validation; it is not part of the scientific evaluator.

The generated provenance manifest is `quantum_dirac_paper/generated/figure_manifest.json`. Human-readable terminal summaries sit beside each experiment. The direct result summary also records the stage-specific replay receipt, ledger hashes, seal hash, freeze and result commits, terminal classification, the known shared π-predicate exception, and non-gating boundary controls. The receipt certifies reproduction of the implemented checkpoint bytes; it does not cure or independently validate the withdrawn predicate.

The complete ledger SHA-256 values are recorded in the generated manifest and the two experiment result summaries; Table 3 prints unambiguous prefixes for reading. The hashes make local evidence tamper-evident; absent an external timestamped deposit they are not themselves an archival guarantee, and they do not replace independent scientific review. The final PDF hash and build-tool versions are written beside the deliverable in `output/pdf`.

A Scenario-level certified distances

Table 4 gives every scenario maximum used in Fig. 3. Values are the serialized outward endpoints of the certified bounds. "Coarse," "middle," and "fine" denote $N = 256, 512, 1024$ for the product walk and $N = 320, 640, 1280$ for the direct QCA.

B Frozen schedules and formula-defined state witnesses

The manuscript generator verifies the SHA-256 digest of each update registration's frozen scenario and state specification before emitting these tables. Write $[t_0, t_1)$ for a schedule interval. The registered mass is piecewise constant, $m(t) = m_j$ on its listed interval, and the potential entries mean

$$V(x,t) = a_j \cos(2\pi h_j x/L + \phi_j)\,, \qquad t \in [t_{0,j}, t_{1,j}), \tag{26}$$

Table 4: All registered scenario-level certified-distance upper bounds.

Architecture / partition	Scenario	Coarse	Middle	Fine
Product walk / development	free light	0.0154366	0.00771499	0.00385708
Product walk / development	free heavy	0.0297087	0.0148466	0.00742234
Product walk / development	mass quench	0.0221470	0.0110755	0.00553848
Product walk / development	potential even	0.0160532	0.00802447	0.00401181
Product walk / confirmation	holdout quench	0.0284341	0.0142083	0.00710284
Product walk / confirmation	holdout potential	0.0300073	0.0149955	0.00749669
Direct QCA / development	free positive	2.3676×10^{-4}	5.9158×10^{-5}	1.4787×10^{-5}
Direct QCA / development	free negative	2.9634×10^{-4}	7.4045×10^{-5}	1.8509×10^{-5}
Direct QCA / development	sign quench	4.0285×10^{-4}	1.0066×10^{-4}	2.5161×10^{-5}
Direct QCA / development	potential H2	0.00218949	0.00108256	5.3983×10^{-4}
Direct QCA / confirmation	holdout pulse	5.6019×10^{-4}	1.3996×10^{-4}	3.4985×10^{-5}
Direct QCA / confirmation	holdout potential H3	0.00275977	0.00137608	6.8787×10^{-4}

with $V = 0$ outside all listed intervals. The product-walk specification has $h_j = 1$ throughout. Table 5 is therefore an exact compact transcription, not a fit to the result ledgers.

Table 5: Exact frozen mass and potential schedules. PW denotes the product walk; DQCA denotes the direct symmetry QCA.

Update / split	Registered scenario	m: value and interval	V: $(h, a, \phi; [t_0, t_1))$
PW / dev.	`DEV-FREE-LIGHT`	$\frac{1}{4}$ $[0, 2)$	0
PW / dev.	`DEV-FREE-HEAVY`	$\frac{1}{2}$ $[0, 2)$	0
PW / dev.	`DEV-QUENCH-UP`	$\frac{1}{4}$ $[0, 1)$; $\frac{1}{2}$ $[1, 2)$	0
PW / dev.	`DEV-POTENTIAL-EVEN`	$\frac{1}{4}$ $[0, 2)$	$(1, \frac{1}{4}, 0; [\frac{1}{2}, \frac{3}{2}))$
PW / conf.	`HOLDOUT-QUENCH-DOWN`	$\frac{1}{2}$ $[0, \frac{3}{4})$; $\frac{1}{4}$ $[\frac{3}{4}, 2)$	0
PW / conf.	`HOLDOUT-POTENTIAL-ODD`	$\frac{1}{2}$ $[0, 2)$	$(1, \frac{1}{4}, \frac{\pi}{2}; [\frac{1}{4}, \frac{5}{4}))$
DQCA / dev.	`DEV-DIRECT-FREE-POS`	$\frac{1}{3}$ $[0, 2)$	0
DQCA / dev.	`DEV-DIRECT-FREE-NEG`	$-\frac{2}{5}$ $[0, 2)$	0
DQCA / dev.	`DEV-DIRECT-SIGN-QUENCH`	$\frac{1}{4}$ $[0, \frac{4}{5})$; $-\frac{1}{3}$ $[\frac{4}{5}, 2)$	0
DQCA / dev.	`DEV-DIRECT-POTENTIAL-H2`	$\frac{2}{5}$ $[0, 2)$	$(2, \frac{1}{8}, 0; [\frac{3}{10}, \frac{13}{10}))$
DQCA / conf.	`HOLDOUT-DIRECT-MASS-PULSE`	$-\frac{1}{5}$ $[0, \frac{7}{20})$; $\frac{3}{5}$ $[\frac{7}{20}, \frac{27}{20})$; $\frac{1}{5}$ $[\frac{27}{20}, 2)$	0
DQCA / conf.	`HOLDOUT-DIRECT-POTENTIAL-H3`	$-\frac{1}{3}$ $[0, 2)$	$(3, \frac{1}{10}, \frac{\pi}{2}; [\frac{1}{4}, \frac{29}{20}))$

Every state was generated from parameters rather than stored amplitudes. Let $\mathsf{N}[v] = v/\|v\|_2$. For $p = (r_0, w, x_0)$, $k_r = 2\pi r/L$, and $r = -4, \ldots, 4$, first form the raw spinor-valued packet and then normalize that packet individually:

$$g_s(r; p, m_0) = e^{-(r-r_0)^2/(2w^2)} e^{-\mathrm{i}k_r x_0} \chi_s(k_r; m_0), \tag{27}$$

$$\psi_s(p; m_0) = \mathsf{N}[g_s(\,\cdot\,; p, m_0)], \qquad s \in \{+, -\}. \tag{28}$$

Here χ_s is the normalized s-energy eigenvector of $k_r\sigma_z + m_0\sigma_x$, fixed by taking its first nonzero component real and positive, and $m_0 = m(0)$ for the scenario. Positive and negative states use $s = +$ and $s = -$, respectively. The implementation then normalizes each composite separately after combining already normalized packets:

$$\psi_{\mathrm{int}}(p_1, p_2, \phi; m_0) = \mathsf{N}\big[\psi_+(p_1; m_0) + e^{\mathrm{i}\phi}\psi_+(p_2; m_0)\big], \tag{29}$$

$$\psi_{\mathrm{Z}}(p, \phi; m_0) = \mathsf{N}\big[\psi_+(p; m_0) + e^{\mathrm{i}\phi}\psi_-(p; m_0)\big]. \tag{30}$$

Table 6 records every exact parameter tuple. For interference rows, p_1 and p_2 are packet tuples; for all other rows, $p = (r_0, w, x_0)$.

C Analytic structure and exact boundaries

For the direct free block, write

$$U_\Delta(k) = aI - \mathrm{i}(b_x\sigma_x + b_z\sigma_z), \qquad \mu = m\Delta, \quad n = \sqrt{1-\mu^2}, \quad a = n\cos(k\Delta), \quad b_x = \mu, \quad b_z = n\sin(k\Delta). \tag{31}$$

Table 6: Exact frozen formula parameters for all state witnesses.

Update / split	Registered state	Type	Exact parameters
PW / dev.	`DEV-STATE-POS`	positive energy	$p = (2, 1, 4)$
PW / dev.	`DEV-STATE-NEG`	negative energy	$p = (-2, 1, 12)$
PW / dev.	`DEV-STATE-INTERFERENCE`	positive interference	$p_1 = (2, 1, 4)$; $p_2 = (-2, 1, 12)$; $\phi = \frac{\pi}{2}$
PW / dev.	`DEV-STATE-ZITTER`	energy mixture	$p = (1, 1, 8)$; $\phi = \frac{\pi}{3}$
PW / conf.	`HOLDOUT-STATE-POS`	positive energy	$p = (-1, \frac{3}{4}, 6)$
PW / conf.	`HOLDOUT-STATE-NEG`	negative energy	$p = (3, \frac{3}{4}, 10)$
PW / conf.	`HOLDOUT-STATE-INTERFERENCE`	positive interference	$p_1 = (1, \frac{3}{4}, 2)$; $p_2 = (-3, \frac{3}{4}, 9)$; $\phi = \frac{\pi}{5}$
PW / conf.	`HOLDOUT-STATE-ZITTER`	energy mixture	$p = (-2, \frac{3}{4}, 3)$; $\phi = -\frac{\pi}{4}$
DQCA / dev.	`DEV-DIRECT-STATE-POS`	positive energy	$p = (\frac{3}{2}, \frac{5}{6}, 3)$
DQCA / dev.	`DEV-DIRECT-STATE-NEG`	negative energy	$p = (-\frac{5}{2}, \frac{2}{3}, 11)$
DQCA / dev.	`DEV-DIRECT-STATE-INTERFERENCE`	positive interference	$p_1 = (\frac{5}{2}, \frac{4}{5}, 2)$; $p_2 = (-\frac{3}{2}, \frac{4}{5}, 13)$; $\phi = \frac{2\pi}{7}$
DQCA / dev.	`DEV-DIRECT-STATE-ZITTER`	energy mixture	$p = (\frac{1}{2}, \frac{4}{5}, 7)$; $\phi = \frac{2\pi}{5}$
DQCA / conf.	`HOLDOUT-DIRECT-STATE-POS`	positive energy	$p = (-\frac{3}{2}, \frac{7}{10}, 5)$
DQCA / conf.	`HOLDOUT-DIRECT-STATE-NEG`	negative energy	$p = (\frac{5}{2}, \frac{7}{10}, 13)$
DQCA / conf.	`HOLDOUT-DIRECT-STATE-INTERFERENCE`	positive interference	$p_1 = (\frac{1}{2}, \frac{3}{4}, 1)$; $p_2 = (-\frac{5}{2}, \frac{3}{4}, 10)$; $\phi = -\frac{\pi}{6}$
DQCA / conf.	`HOLDOUT-DIRECT-STATE-ZITTER`	energy mixture	$p = (-\frac{1}{2}, \frac{3}{4}, 4)$; $\phi = \frac{3\pi}{7}$

Let $\rho = \sqrt{b_x^2 + b_z^2}$ and $\omega = \operatorname{atan2}(\rho, a)$ on the principal branch. Because $a^2 + \rho^2 = 1$, the principal effective generator is exactly $H_{\text{eff}} = \frac{\omega}{\Delta\rho}(b_x\sigma_x + b_z\sigma_z)$, with its continuous $\rho \to 0$, $a \to +1$ value understood in the registered branch-safe neighborhood of the identity. Expanding ω/ρ and the two axis components gives Eqs. (18) and (19). An independent matrix-log fixture reconstructs the same generator from a power series without calling the axis–angle implementation. Halving Δ produces the predeclared 1/16 corrected-generator residual ratio and 1/8 one-step propagator ratio on disclosed nonregistered fixtures.

The global direct spectrum follows from

$$\det U_\mu(q) = 1, \qquad \operatorname{tr} U_\mu(q) = 2n \cos q. \tag{32}$$

For $0 < |\mu| < 1$, group velocity obeys

$$|v(q)| \le n = \sqrt{1 - \mu^2} \le 1, \tag{33}$$

consistent with radius-one causal support. At $|\mu| = 1$ the bands are flat at $\pm\pi/2$ and group velocity is zero. For $|\mu| > 1$, n is imaginary and the displayed family is not unitary. The registered direct margin $|\mu| \le 3/100$ at the coarsest step is therefore strict.

The exact hard-core collision choice appears in Eq. (15). For a number-preserving one-particle unitary V, canonical fermionic second quantization acts on double occupation by $\det V$. Both the edge-scatter one-particle block and the routing SWAP have determinant -1, while the declared qubit gates use phase $+1$. This algebraic distance of 2 is the post-primary collision control reported in Sec. 5.

D Interpretation of confirmation

The primary comparison is a deterministic, certified statement uniform over the finite input sector. Consequently, the confirmation partition does not estimate a frequentist population effect. Its evidential role is procedural and interventional: exact schedules and state formulas unused in update construction or development assessment are opened once, propagated by the sealed code, and required to pass without replacing any map, threshold, or candidate byte. Reporting scenario and witness counts should therefore not be read as a sample-size claim. The correctly implemented members of the registered finite test suite passed, and the evolution certificate covers every normalized state in $\mathcal{K}_{\text{in}}$ for those scenarios. The empirical π-sector member is excluded from that sentence because its empty-set reducer is withdrawn under Sec. 4.5.

References

[1] Pablo Arrighi, Stefano Facchini Forets, and Vincent Nesme. The Dirac equation as a quantum walk: higher dimensions, observational convergence. *Journal of Physics A: Mathematical and Theoretical*, 47(46):465302, 2014. doi: 10.1088/1751-8113/47/46/465302.

[2] Alessandro Bisio, Giacomo Mauro D'Ariano, and Alessandro Tosini. Quantum field as a quantum cellular automaton: The Dirac free evolution in one dimension. *Annals of Physics*, 354:244–264, 2015. doi: 10.1016/j.aop.2014.12.016.

[3] Giacomo Mauro D'Ariano and Paolo Perinotti. Derivation of the Dirac equation from principles of information processing. *Physical Review A*, 90(6):062106, 2014. doi: 10.1103/PhysRevA.90.062106.

[4] Alessandro Bisio, Giacomo Mauro D'Ariano, and Alessandro Tosini. Dirac quantum cellular automaton in one dimension: Zitterbewegung and scattering from potential. *Physical Review A*, 88(3):032301, 2013. doi: 10.1103/PhysRevA.88.032301.

[5] Terence C. Farrelly and Anthony J. Short. Causal fermions in discrete spacetime. *Physical Review A*, 89(1):012302, 2014. doi: 10.1103/PhysRevA.89.012302.

[6] Leonard Mlodinow and Todd A. Brun. Quantum field theory from a quantum cellular automaton in one spatial dimension and a no-go theorem in higher dimensions. *Physical Review A*, 102(4):042211, 2020. doi: 10.1103/PhysRevA.102.042211.

[7] János K. Asbóth and Hideaki Obuse. Bulk-boundary correspondence for chiral symmetric quantum walks. *Physical Review B*, 88(12):121406, 2013. doi: 10.1103/PhysRevB.88.121406.

[8] Dogukan Bakircioglu, Pablo Arnault, and Pablo Arrighi. Fermion doubling in quantum cellular automata, 2025. Preprint, arXiv:2505.07900.

[9] Chaitanya Gupta and Anthony J. Short. Fermion doubling in Dirac quantum walks. *Physical Review A*, 114(1):012208, 2026. doi: 10.1103/kkz3-fthm.

[10] Chaitanya Gupta and Anthony J. Short. The Dirac vacuum in discrete spacetime. *Quantum*, 9:1845, 2025. doi: 10.22331/q-2025-09-03-1845.